



Course Numerical Computing

Answer Sheet No. 34756

Student's Name _____

Signature _____

Roll No. _____

in BCS-2

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QUESTION-01

* Newton Backward Interpolation Formula:

$$Y_p = Y_n + p \nabla Y_n + \frac{p(p+1)}{2!} \nabla^2 Y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 Y_n \quad \text{--- (1)}$$

$$p = \frac{x - x_n}{h} \quad \text{--- (2)}$$

⇒ Chain rule: $\frac{dy}{dx} = \frac{dy}{dp} \times \frac{dp}{dx}$

⇒ $\frac{dp}{dx}$ using (2) ⇒ $\boxed{\frac{dp}{dx} = \frac{1}{h}}$

$$\begin{aligned} & p^2 + p \\ & \cdot \frac{p(p^2 + 2p + 2)}{3!} \\ & \cdot \frac{p(p^3 + 3p^2 + 2p)}{3!} \end{aligned}$$

= $\frac{dy}{dp}$ using (1) ⇒ $\frac{dy}{dp} = \nabla Y_n + \frac{(2p+1)}{2!} \nabla^2 Y_n + \frac{(3p^2 + 6p + 2)}{3!} \nabla^3 Y_n$

$$\boxed{\frac{dy}{dx} = \frac{1}{h} \left(\nabla Y_n + \frac{(2p+1)}{2!} \nabla^2 Y_n + \frac{(3p^2 + 6p + 2)}{3!} \nabla^3 Y_n \right)}$$

x	Y	∇Y_n	$\nabla^2 Y_n$	$\nabla^3 Y_n$
75	38.2			
90	43.2	5		
105	40.9	-2.3	-7.3	
120	37.7	-3.2	-0.9	6.4

To find $f'(120)$; $x = 120$, $x_n = 120$; $p = 0$,
 $h = 15$

Part No.

$$\left[\frac{dy}{dx} \right]_{x=x_0} = \left(\Delta y_0 + \frac{1}{2} \Delta^2 y_0 + \frac{2}{6} \Delta^3 y_0 \right) \frac{1}{15}$$

$$= \left((-3.2) + \frac{1}{2} (-0.9) + \frac{1}{3} (6.4) \right) \frac{1}{15}$$

$$= \boxed{-1.5167} \text{ Ans. } \boxed{-0.1011} \text{ Ans.}$$

QUESTION - 02

f(25)

X	Y	ΔY	Δ ² Y	Δ ³ Y
20	2765			
23	3261	496		
26	3454	193	-303	
29	3929	475	282	585

$$p = \frac{x - x_0}{h} ; h = 3$$

$$p = \frac{25 - 23}{3} = \frac{2}{3}$$

Using Bessels:

$$y_p = \frac{y_0 + y_1}{2} + \frac{p - 0.5}{2} \Delta y_0 + \frac{p(p-1)}{2} \left[\frac{\Delta^2 y_0 + \Delta^2 y_1}{2} \right]$$

$$+ \frac{p(p-0.5)(p-1)}{6} \Delta^3 y_0$$

$$= \frac{3261 + 3454}{2} + \left(\frac{2}{3} - 0.5 \right) (193) + \left(\frac{2}{3} \right) \left(\frac{2}{3} - 1 \right) \left[\frac{282 - 303}{2} \right] + \frac{\left(\frac{2}{3} \right) \left(\frac{2}{3} - 0.5 \right) \left(\frac{2}{3} - 1 \right) (585)}{6}$$

$$= \boxed{3388.33} \text{ Ans.}$$

$$= \boxed{3375.33} \text{ Ans.}$$

QUESTION-03

$$\int_0^1 f(x) dx$$

X	Y
$x_0 = 0$	$1 = y_0$
$x_1 = 0.25$	$\beta = y_1$
$x_2 = 0.5$	$2.5 = y_2$
$x_3 = 0.75$	$\beta = y_3$
$x_4 = 1$	$2 = y_4$

$$\therefore \text{Simpson's rule: } h \left[y_0 + y_n + 4(y_1 + y_3 + \dots) + 2(y_2 + y_4 + \dots) \right]$$

$$\frac{0.25}{3} [1 + 2 + 4(\beta + \beta) + 2(2.5)] = 2.75$$

$$\frac{1}{12} [3 + 4(2\beta) + 5] = 2.75$$

$$3 + 8\beta + 5 = 33$$

$$8\beta = 33 - 8$$

$$8\beta = 25$$

$$\boxed{\beta = 3.125} \text{ Ans.}$$

QUESTION-04

→ approx less than 1×10^{-5}

$$\int_0^1 e^x dx \quad ; a=0, b=1$$

⇒ Trapezoidal Rule:

$x_0 \dots x_n$

$$\int_{x_0}^{x_n} f(x) dx \approx \frac{h}{2} [y_0 + y_n + 2(y_1 + y_2 + \dots + y_{n-1})]$$

$$E_T = \frac{-h^2(b-a)}{12} f''(\xi) \quad \therefore h = \frac{b-a}{n}$$

$$f(x) = e^x$$

$$f'(x) = e^x$$

$$f''(x) = e^x$$

$$E_T = \frac{-h^2(1)(e^x)}{12}$$

$$E_T = \frac{-h^2 e^x}{12} \quad , \quad |E_T| = \frac{h^2 e^x}{12}$$

$$\frac{h^2 e^x}{12} < |E_T| < \frac{h^2 e^x}{12}$$

$$\frac{h^2 e^0}{12} < |E_T| < \frac{h^2 e^1}{12}$$

$$\frac{h^2}{12} \leq 1 \times 10^{-5}$$

$$h^2 \leq 1.2 \times 10^{-4}$$

$$h \approx \sqrt{1.2 \times 10^{-4}}$$

$$h \approx 0.011$$

$$\frac{h^2 e}{12} \leq 1 \times 10^{-5}$$

$$h^2 e \leq 1.2 \times 10^{-4}$$

$$h^2 \leq \frac{1.2 \times 10^{-4}}{e}$$

$$h^2 \leq (4.41 \times 10^{-5})$$

$$h \approx 6.64 \times 10^{-3}$$

$$n = \frac{b-a}{h}$$

$$n = \frac{1}{0.011}$$

$$n = \frac{1}{6.64 \times 10^{-3}}$$

$$n = 90.9$$

$$n = 91$$

$$n = 150.602$$

$$n = 151$$

$$H = 0.01$$

$$N = 91$$

$$10$$